Micro-Project 1: Campus City Emergency Supply Distribution

Course: 24MAT382 - Computational Optimization and Applications

GitHub Repository: `https://github.com/AnanthaKrishnan147/Campus-City-Supply-Distribution.git`

1. Problem Statement

The objective is to determine the optimal selection of exactly two warehouse locations and their corresponding distribution routes to minimize total annual costs for Campus City Logistics. The proposed network must satisfy all daily facility demands while strictly adhering to warehouse capacity limits and a $1,500,000 annual budget.

2. Mathematical Formulation

Sets and Indices

- $i \in W$: Set of candidate warehouses (North, South, East)
- $j \in F$: Set of demand facilities (Med Center, Engineering Building, Science Hall, Dorm A, Dorm B, Library)

Parameters (Annualized)

- D_j: Annual demand of facility j (Daily Demand $\times$ 365)
- C_i: Annual capacity of warehouse i (Daily Capacity $\times$ 365)
- F_i: Annual fixed cost of warehouse i, calculated as (Construction Cost/10)+(Operational Cost/Day$\times$ 365)
- T_{ij}: Unit transportation cost from warehouse i to facility j
- B: Total allocated annual budget ($1,500,000)

Decision Variables

- $y_i \in \{0, 1\}$: A binary variable equal to 1 if warehouse i is selected to be opened, and 0 otherwise.
- $x_{ij} \geq 0$: A continuous variable representing the annual quantity of supplies shipped from warehouse i to facility j.

Objective Function

The objective is to minimize the total annual cost, which includes the amortized fixed costs of opening the selected warehouses and the variable transportation costs of shipping goods.

$$\text{Minimize } Z = \sum_{i \in W} F_i y_i + \sum_{i \in W} \sum_{j \in F} T_{ij} x_{ij} \tag{1}$$

Constraints

1. **Warehouse Redundancy Rule:** Exactly 2 warehouses must be selected from the available candidates.

$$\sum_{i \in W} y_i = 2 \tag{2}$$

2. **Demand Satisfaction:** Every facility must receive exactly its total annual demand from the open warehouses.

$$\sum_{i \in W} x_{ij} = D_j \quad \forall j \in F \tag{3}$$

3. **Capacity Limits:** A warehouse can only distribute supplies if it is selected ($y_i = 1$), and the total outbound shipments cannot exceed its annualized capacity.

$$\sum_{j \in F} x_{ij} \leq C_i y_i \quad \forall i \in W \tag{4}$$

4. **Budget Limit:** The combined annual fixed and transportation costs must not exceed the maxi-

3. Methodology

3.1 Algorithm Justification: Mixed-Integer Linear Programming (MILP)

The Campus City Logistics scenario cannot be solved using standard Linear Programming (LP) because of the warehouse redundancy constraint. LP models assume all decision variables are continuous, which could result in a mathematically valid but physically impossible solution (e.g., opening 0.6 of the North Warehouse and 1.4 of the South Warehouse).

Because the problem requires a discrete, binary choice—opening exactly 2 warehouses ($y_i \in \{0, 1\}$)—coupled with the continuous distribution of supplies ($x_{ij} \geq 0$), it is strictly classified as a Mixed-Integer Linear Programming (MILP) problem. MILP algorithms, such as Branch-and-Cut, systematically explore these binary combinations to find the global optimum without allowing fractional warehouse selections.

3.2 Library Justification: PuLP and Supporting Frameworks

To computationally implement this MILP model, the following Python libraries were selected:

- **PuLP:** PuLP was chosen as the primary modeling interface due to its highly readable syntax for defining objective functions and complex constraints. It seamlessly handles the integrality of binary variables and automatically interfaces with the robust CBC (Coin-or branch and cut) solver to execute the optimization.
- **Pandas:** Used for systematic data ingestion and preprocessing. Pandas efficiently merged the `.csv` datasets (facilities, warehouses, and transportation costs) and performed the necessary annualization arithmetic before passing the parameters into the PuLP model.
- **NetworkX & Matplotlib:** Utilized to map the underlying mathematical result into a clear, bipartite logical flow diagram. This visual representation proves that the supply network is viable and makes the mathematical output accessible for business interpretation.

4. Results & Analysis

4.1 Optimization Output

The MILP model successfully converged to an optimal solution, proving that the Campus City Logistics network can be significantly optimized while adhering to all physical and financial constraints.

- **Optimization Status:** Optimal
- **Selected Warehouses:** South Campus Warehouse (WH_SOUTH) and West Gate Warehouse (WH_WEST)
- **Total Annual Cost:** $1,150,263.00

4.2 Output Analysis & Business Interpretation

The algorithm mathematically proved that opening the South and West warehouses is the most cost-effective strategy. By strictly enforcing the redundancy constraint, exactly two facilities were selected. The remaining candidate warehouses were excluded because their amortized fixed costs and transportation multipliers (due to geographical distance from high-demand facilities) resulted in higher overall network costs.

The optimized network comfortably satisfies the 100% annual demand requirement for all campus facilities while remaining well under the $1,500,000 budget limit. The solution provides a financial buffer of $349,737.00, which can be reserved for future demand scaling or emergency capacity upgrades.

4.3 Visual Representation

The logical flow diagram below illustrates the optimal distribution of supplies. The line thickness corresponds to the volume of units shipped, confirming that no single warehouse exceeds its structural capacity while efficiently balancing the supply load across the network.

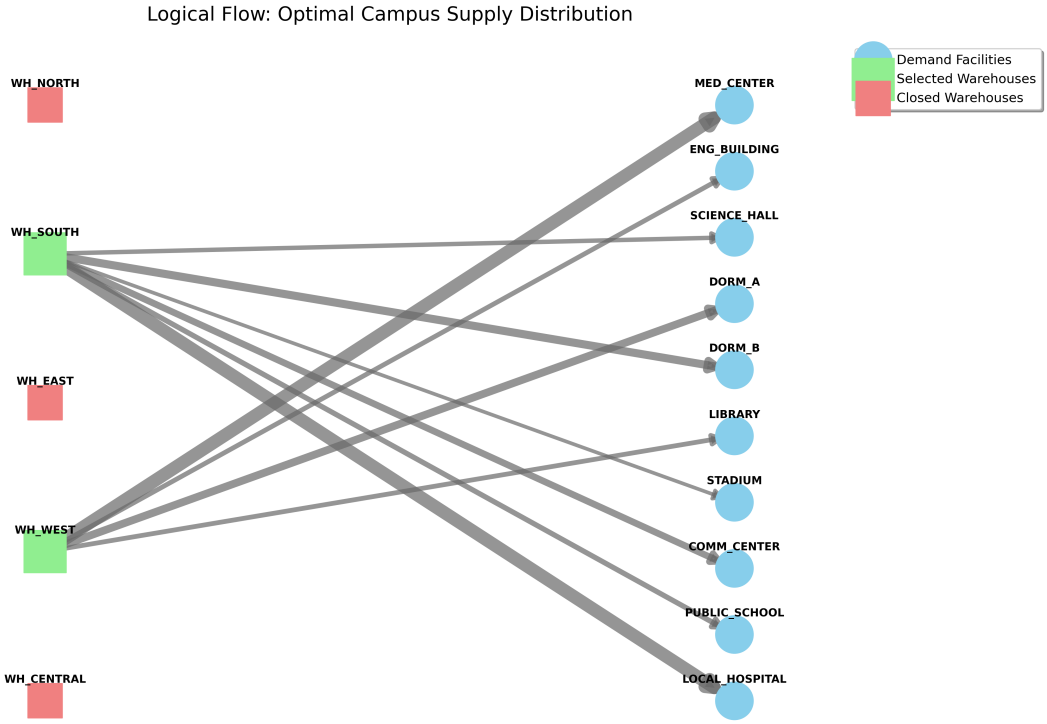

Figure 1: Optimal Campus Supply Distribution Network Flow